

Functional & Performance Testing Template

Model Performance Test

Date	30 June 2026
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	4 Marks

Test Scenarios & Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	Soil Parameter Input Validation (N, P, K, Temperature, Humidity, pH, Rainfall)	Enter valid and invalid values for each soil/environmental input field (e.g., letters in numeric fields, empty fields)	Valid inputs accepted; appropriate error messages shown for invalid or missing inputs	Input validation working correctly; valid soil values accepted, errors shown for invalid entries	Pass
FT-02	Numeric Range Validation (N: 0-140, P: 0-145, K: 0-205, pH: 0-14, Rainfall: 0-300 mm)	Enter numbers within and outside each valid agronomic range for N, P, K, pH, Temperature, Humidity, and Rainfall	Accepts valid agronomic values; shows error for out-of-range inputs	Numeric ranges validated correctly; out-of-range values rejected with error messages	Pass
FT-03	Crop Recommendation Generation (Logistic Regression Model)	Provide complete soil and environmental inputs (N, P, K, Temperature, Humidity, pH, Rainfall) and click "Recommend"	System recommends the most suitable crop (e.g., Rice, Maize, Wheat) based on ML model output	Crop recommendation generated successfully; model predicted correct crop based on inputs	Pass
FT-04	ML Model and Backend Integration Check	Verify that the saved ML model loads correctly and the Python backend responds to requests	Backend responds successfully; model loads without errors and returns crop predictions	ML model loaded; backend responded to frontend requests without errors	Pass

		from the HTML frontend			
PT-01	Crop Recommendation Response Time Test	Submit full soil/environmental inputs and use a timer to measure time from clicking Recommend to displaying the crop result	Should be under 3 seconds	Crop recommendation displayed within 2 seconds (96% accuracy model)	Pass
PT-02	Concurrent User / Model Prediction Speed Test	Simulate multiple simultaneous prediction requests with varied soil/environmental inputs and check response consistency	Model should not slow down or return incorrect predictions under concurrent requests	All concurrent requests processed correctly; no slowdown or incorrect predictions	Pass
PT-03	Agricultural Dataset Upload Load Test (e.g., CSV files)	Upload multiple agricultural CSV dataset files and check processing time and application stability	System processes all uploads smoothly without crashing or data loss	All CSV files processed smoothly without crash or data loss	Pass